

# Expression quantitative trait loci (eQTL)

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**DNA**

ACTGACCTAGATCAGTGTAGCGATCGTATACGAGACCGATTTCATCGGCAT



**transcription**

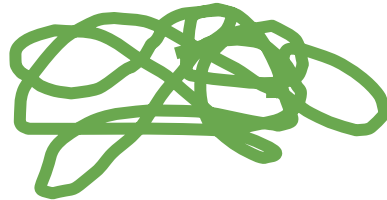
**RNA**

AUCAGUCGAUCACCGAU

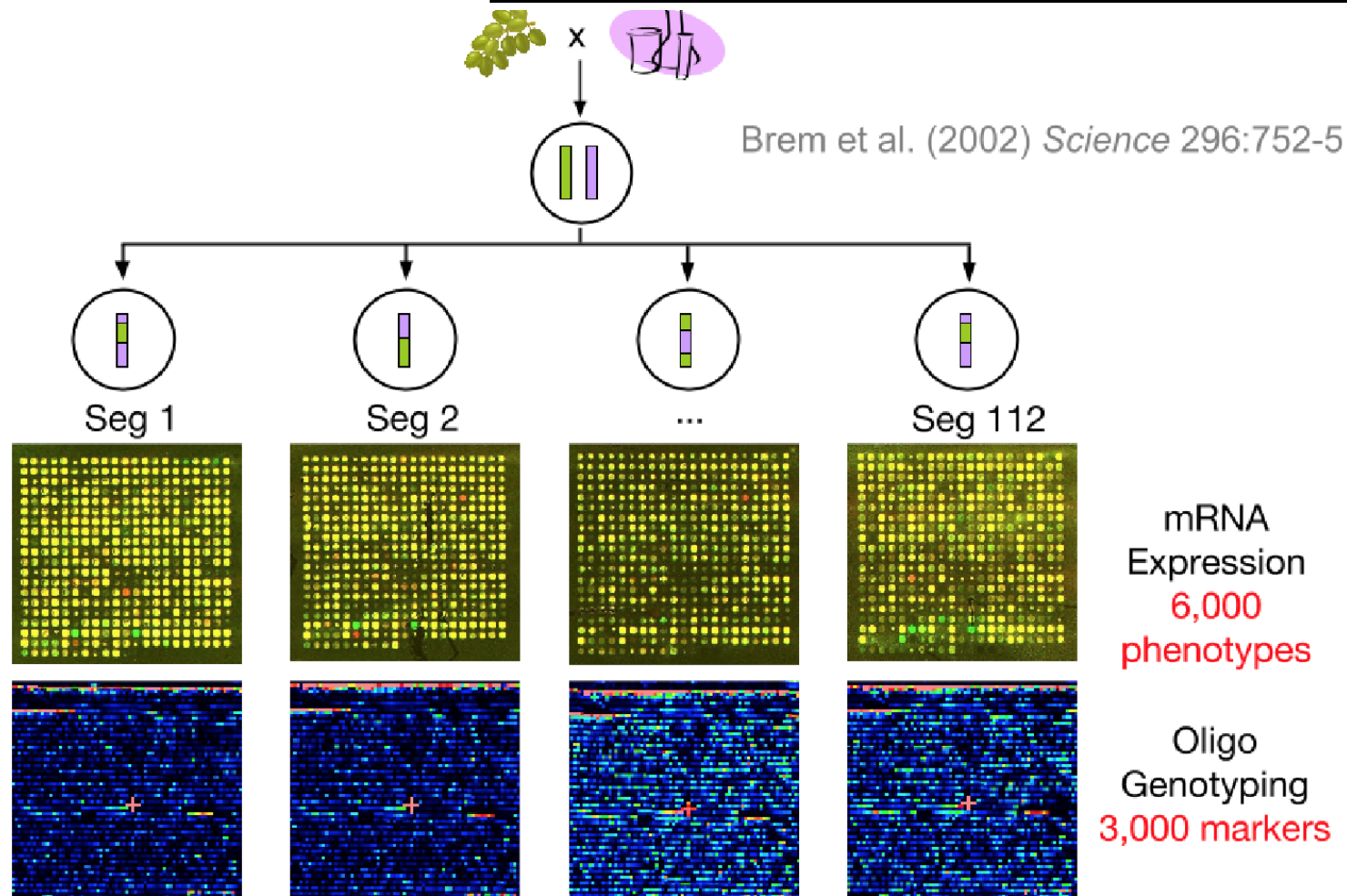


**translation**

**protein**



# Study design



## Chromosomes

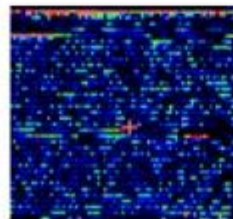
n=16 in yeast



...ACTACAGAC**CCA**ACCTGGGGATGCCGCGGAACAGGGAATAGCC...



Marker (SNP)  
Position



Oligo  
Genotyping  
3,000 markers

Data

$G_1, \dots, G_{1226}$

$G_i = (g_{i1}, \dots, g_{i112})$

## Chromosomes

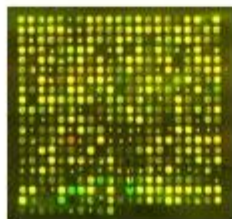
n=16 in yeast



...ACTACAGACC**CAACCTGGGG**ATGCCGCGGAACAGGGAATAGCC...



Trait (Gene)  
Position

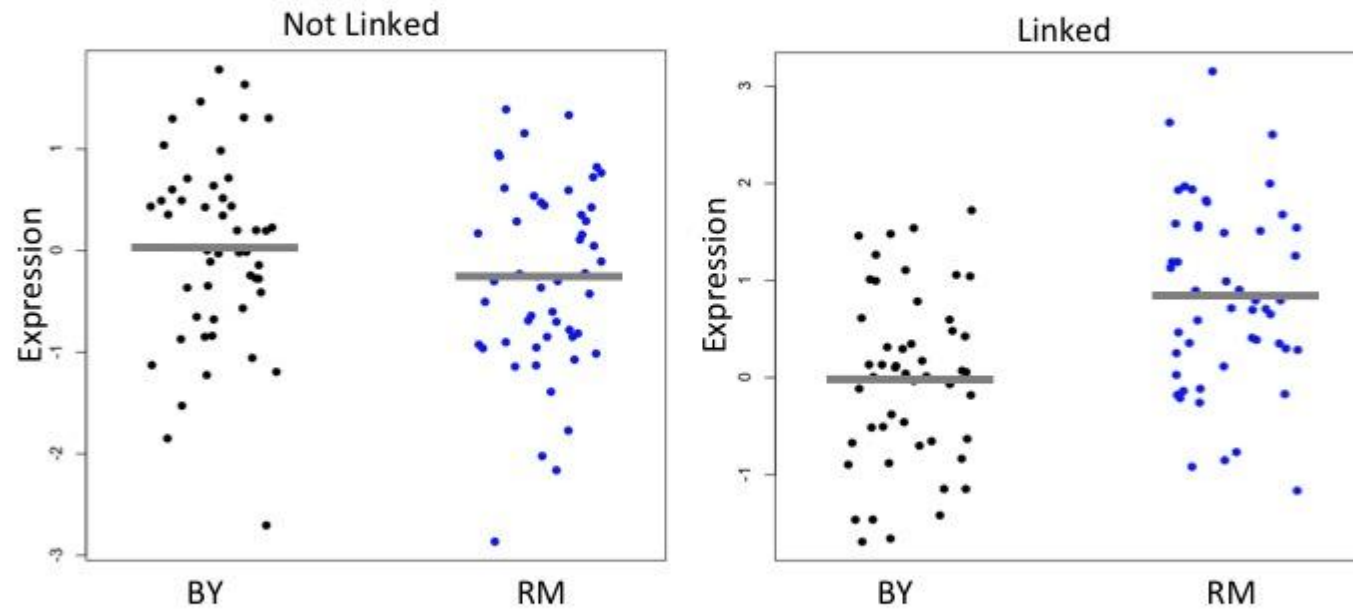


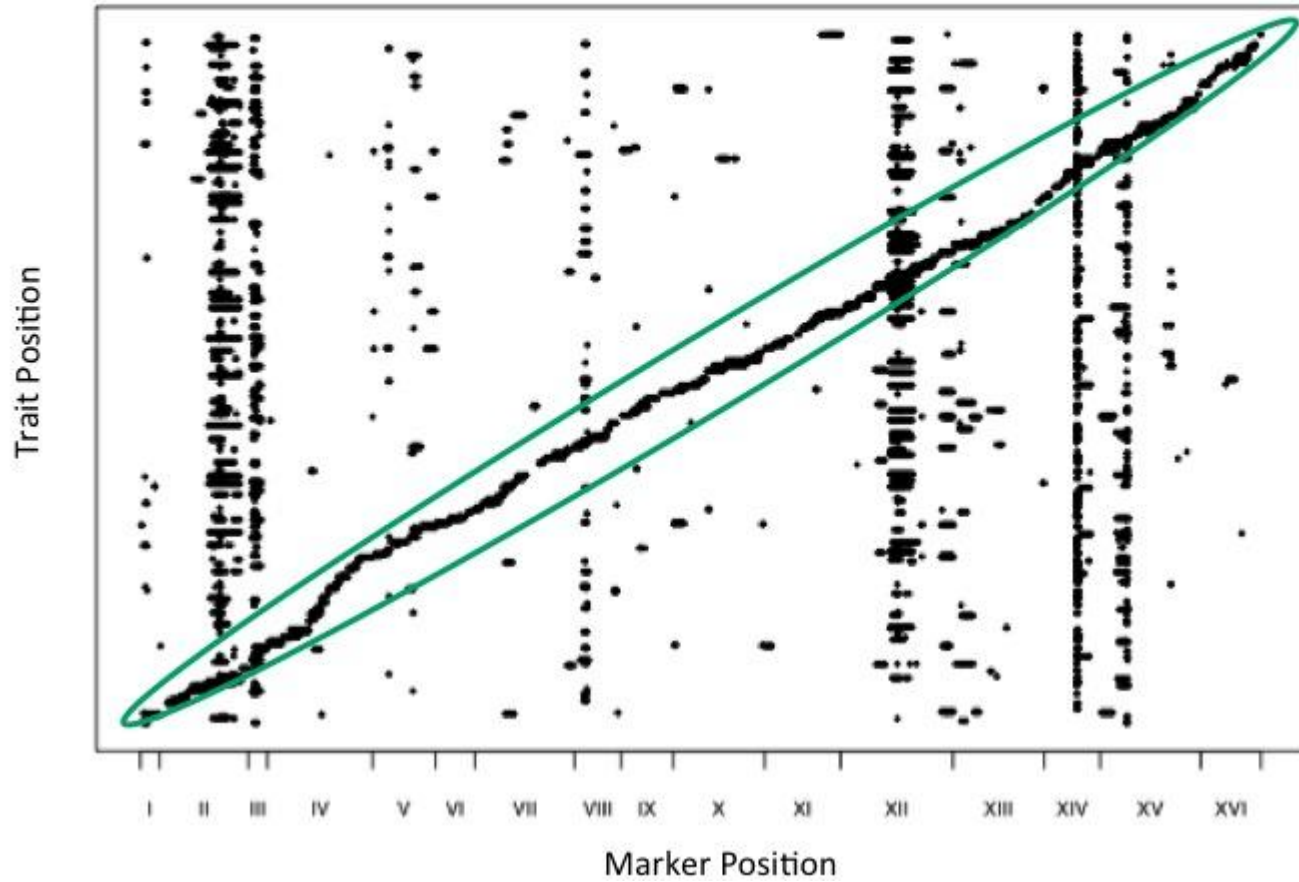
mRNA  
Expression  
6,000  
phenotypes

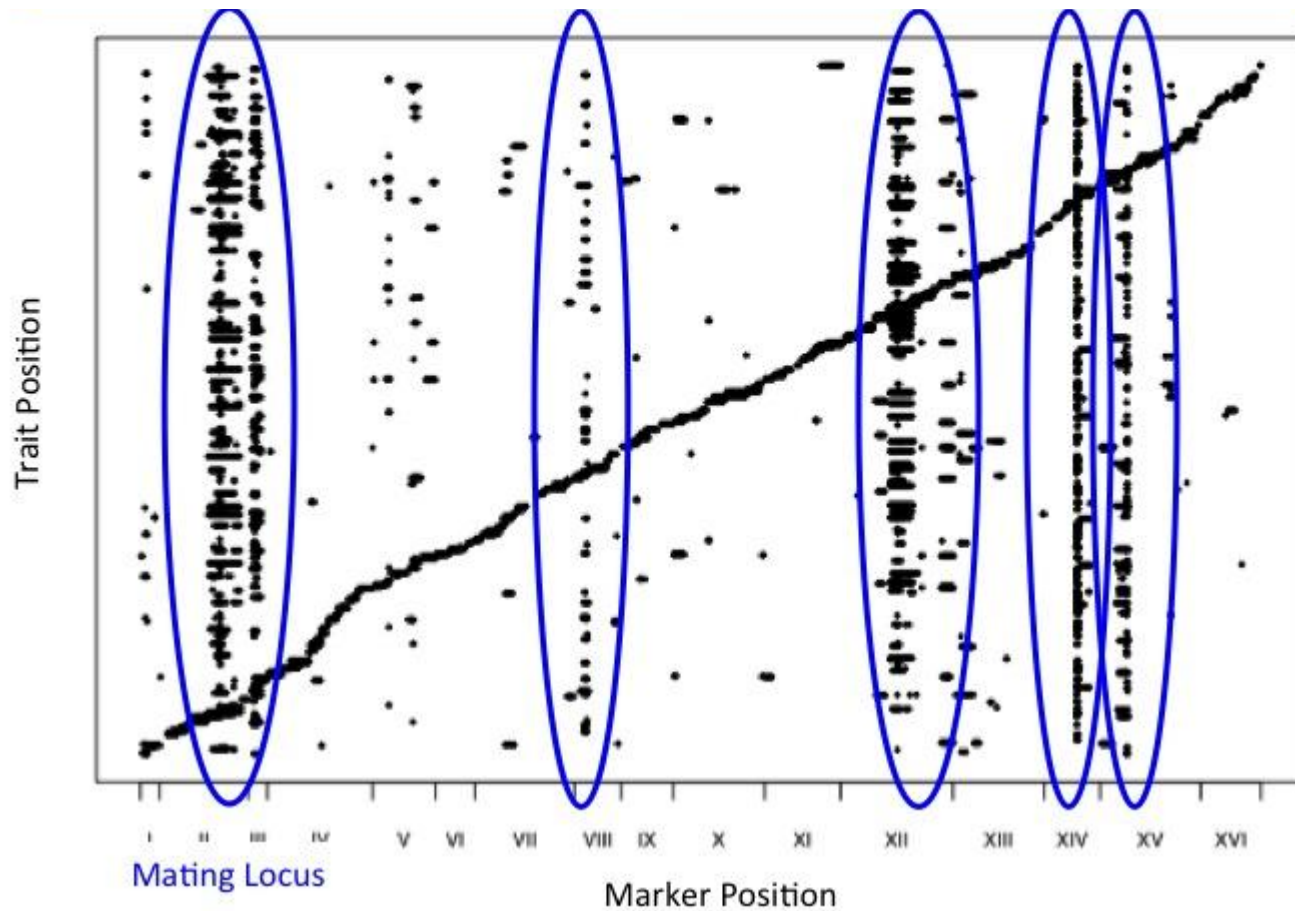
Data

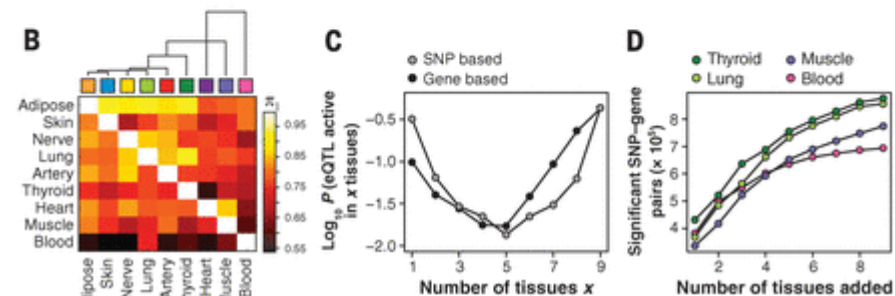
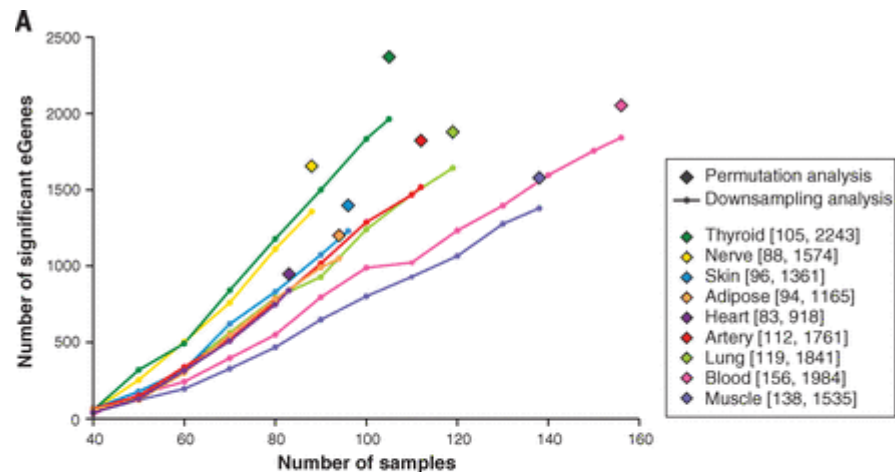
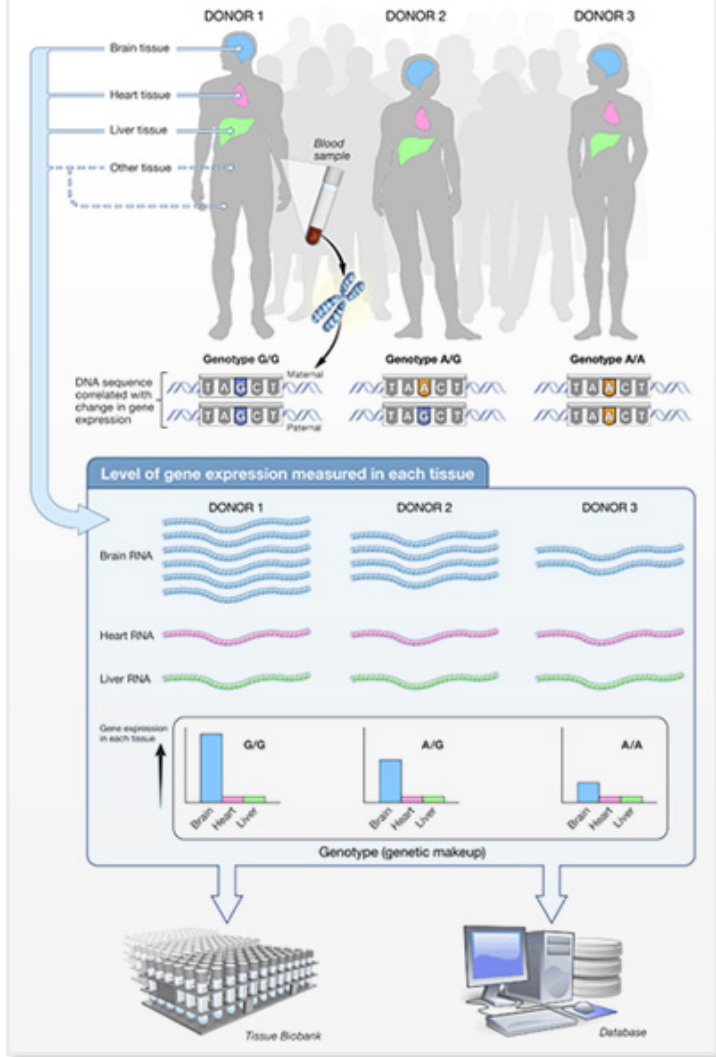
$$X_1, \dots, X_{6216}$$
$$X_i = (x_{i1}, \dots, x_{i6216})$$

# Statistical model









# Notes and further reading

- cis-eQTL are usually more believable than trans-eQTL
- There are many potential confounders
  - Population stratification, batch effects, sequence artifacts
- An excellent review
  - <http://www.nature.com/nrg/journal/v16/n4/abs/nrg3891.html>